Do Small Local LLM Rerankers Help Scientific Paper Recommendation?

An Audit and Re-evaluation with Citation and Human Expertise Labels

Sena Oz
Boğaziçi University
Istanbul, Türkiye

Abstract

Small LLMs that run locally are an attractive second stage for scientific paper recommenders. We built such a recommender for researchers at Boğaziçi University and asked whether reranking embedding-retrieved papers with Llama improves it. Our first evaluation said it did. We audit that evaluation, find nine problems, chief among them that the cited papers were always shown to the LLM first, and re-evaluate on a rebuilt, leak-free citation benchmark (118 queries, 13 methods and a larger Llama) and on expertise ratings from 58 researchers. We find that listwise LLM rerankers are sensitive to candidate order in a model-dependent direction: Llama 3.2 3B favours the first candidates and Llama 3.1 8B the last, so a single fixed order inflates the 3B model's score and can lift the 8B model above dense retrieval. Under shuffled order, no LLM reranker, second stage or citation-graph signal we tested beats Qwen3-Embedding-8B on both label sources, and a 22M-parameter encoder comes within about four R-precision points of it. We recommend evaluating LLM rerankers under several candidate orders and checking citation-derived labels against human judgments.

Keywords

scientific paper recommendation, LLM reranking, position bias, evaluation

1 Introduction

Researchers miss relevant work from institutions they do not follow. Our system monitors new papers from a set of institutions and, for each one, lists the Boğaziçi researchers who work on its topic (Figure 1a). It retrieves candidates with an embedding model and was designed to rerank them with a small LLM that runs on local hardware, an attractive choice for a university system that would rather not depend on a paid, hosted model. Whether that second stage earns its cost is the question of this paper.

Our first evaluation answered it in the LLM's favour: on a citation benchmark, listwise Llama 3.2 3B reached MRR 0.84 against 0.51 for Qwen3-Embedding-8B. Auditing our own code, we found nine problems with that evaluation (Section 3); this paper is a reproducibility study of our own claim. The most consequential for the LLM was that the cited papers were always shown first; we show below that candidate order alone reproduces its score, while its lead over the embeddings also depended on an indexing error that gave positive candidates other papers' text and so degraded every other method. We therefore rebuilt the evaluation and tested every conclusion on a second, independent label source: the self-reported reviewing expertise of 58 researchers [34].

Our contributions are:

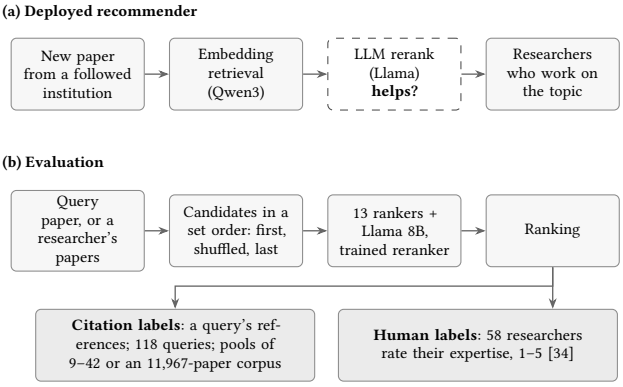


Figure 1: (a) Our recommender retrieves Boğaziçi researchers for a new paper with an embedding model and was designed to rerank them with a small LLM; we ask whether that stage helps. (b) We present the same candidates to every ranker in a controlled order and score the ranking on two independent label sources.

- an audit that traces our first result to nine problems, listed as a checklist for evaluating LLM rankers (Section 3);
- a leak-free citation benchmark for paper recommendation, with controlled candidate order, two kinds of negatives and an 11,967-paper corpus whose co-citation graph obeys explicit, unit-tested leakage rules (Section 4);
- evidence that listwise LLM rerankers are sensitive to candidate order in a direction that depends on the model, Llama 3.2 3B favouring the first candidates and Llama 3.1 8B the last, on citation and human labels alike (Section 5);
- a comparison of 13 methods on citation labels and eight on human labels, which order the shared methods similarly (Section 6);
- negative results: neither a second stage nor citation-graph signals improve a strong embedding model on both label sources (Sections 7 and 8).

Code, data and per-query results are available at https://github.com/senaoz/SWE-599.

2 Related Work

LLMs as rerankers. LLMs have been used to rerank retrieved candidates pointwise, by scoring each candidate on its own [28]; pairwise, by comparing two at a time [25]; setwise [44]; and listwise, by returning an order for a whole list [18, 35], which has been distilled into open models [23, 24]. Listwise prompting with a sliding window over long lists [35] is the setting closest to our cascade.

In recommendation, LLMs have been studied as zero-shot rankers of candidate items [6, 12]. Most of this work uses large hosted or fine-tuned models; we ask what a small local model, the kind a university system can afford to run, adds to an embedding retriever.

Position bias. LLM outputs depend on the position of information in the prompt: models use information at the start and end of a long context better than in the middle [17], few-shot predictions drift toward recent examples [41], LLM judges prefer answers by their position [37, 43], and multiple-choice answers depend on option order [22, 42]. For ranking, Tang et al. [36] reduce the effect by aggregating rankings over shuffled inputs, and Hou et al. [12] report that the order of candidates affects LLM recommenders. We add two observations: the direction of the bias changed between two Llama models of different size and release (3.2 3B and 3.1 8B), and a single fixed order in the evaluation can make a reranker look better than dense retrieval.

Scientific documents. Some encoders learn document embeddings from citations, as SPECTER [5] and SPECTER2 [32] do; others are general sentence encoders [26, 38] or large embedding models [31, 40], compared on benchmarks such as MTEB [21] and SciRepEval. Citation recommendation uses a paper's references as relevance labels [3, 7], a practice that surveys of research-paper recommenders also document [2]. We use such labels too, but test every conclusion against human judgments.

Expert finding and reviewer assignment. Matching papers to people is studied as expertise retrieval [1] and reviewer assignment [4, 20]. The gold standard of Stelmakh et al. [34], in which researchers rated their own expertise, is our second label source; its direction (papers for one researcher) is the reverse of our system's (researchers for one paper).

Citation graphs and evaluation practice. Co-citation [33] and bibliographic coupling [13] are classic relatedness signals, and corpus graphs have been used to expand dense retrieval results [15, 19]. Evaluating them on citation labels risks leakage, which we control with explicit rules. Our audit of our own first benchmark joins a line of work showing how weak baselines and flawed evaluation inflate reported progress in recommendation and retrieval [8, 9, 16].

3 Auditing the First Evaluation

Our first benchmark ranked each query's cited references among sampled distractors and reported MRR 0.84 for listwise Llama 3.2 3B against 0.51 for Qwen3-Embedding. Auditing it, we found nine problems, which we list as a checklist for LLM ranking evaluations. Each item below was confirmed from the code or the stored outputs; per-query effects of F4 and F6 could not be recomputed because the original candidate pools no longer exist.

F1 Candidates were passed to every LLM method as positives followed by negatives.
F2 The prompt's format example, `[3, 1, 7, . . . ]`, pointed at a positive.
F3 A parse or HTTP failure returned the input order, putting positives first.
F4 Negatives lacked the concept field that positives had, so the two groups differed in format.

Table 1: Evaluation settings. Binary labels mark cited papers; τ loss is the weighted Kendall-τ loss of Stelmakh et al. [34]. Citation settings report the test split; tuning used a separate dev split.

Setting	Units	Candidates	Label	Metric
Citation pools	118 queries	9–42	binary	R-precision
Full corpus	118 queries	11,967	binary	recall@100
Human labels	58 researchers	3–10	1–5	τ loss

F5 The drop of the two-stage cascade was attributed to first-stage recall, which it was not.
F6 A label-versus-position indexing error gave positive candidates the text of other papers, pushing every non-LLM method towards chance.
F7 nDCG was normalized by the hits found rather than by the ideal ranking.
F8 Negative sampling was seeded with Python's per-process salted `hash()`, so the pools could not be reproduced.
F9 SPECTER2 was encoded with mean pooling instead of its [CLS] token.

F1–F4 affect only the LLM methods, and F6 every method. The benchmark in Section 4 was rebuilt so that each item is checked in code, and its runs store the order shown to every method. Because the first pools cannot be regenerated (F8), we compare the two versions claim by claim rather than query by query.

4 Evaluation Setup

We compare 13 methods on citation labels, the references of 118 query papers ranked within pools of at most 42 candidates and within an 11,967-paper corpus, and a subset of them on an independent label source, the self-reported reviewing expertise of 58 researchers [34]. Table 1 summarizes the three settings.

4.1 Task and Methods

Given a query, each method ranks its candidates so that relevant papers come first. We compare two lexical methods, BM25 [27] and TF-IDF; four dense encoders scored by cosine similarity, MiniLM (all-MiniLM-L6-v2) [26, 38], SPECTER2 with [CLS] pooling [5, 32], EmbeddingGemma [31] and Qwen3-Embedding-8B [40]; the cross-encoder bge-reranker-base [39]; Llama 3.2 3B Instruct [10] used listwise over the full list, listwise with a sliding window of 10 candidates and step 5 [35], and pointwise; and three combinations, a two-stage cascade in which Llama reranks Qwen3-Embedding's top 10 and two score fusions (Llama + MiniLM, Qwen3 + MiniLM). To test scale, we repeat the listwise experiments with Llama 3.1 8B Instruct, and in the full-corpus and human-label settings we add Qwen3-Reranker-4B [40], a model trained for reranking.

All models run locally: the Llama models and Qwen3-Embedding as 4-bit (Q4_K_M) GGUF files in llama.cpp, the others in sentence-transformers or transformers. On citation labels every method receives the same text, the title, abstract and OpenAlex concepts of each paper, truncated to 600 characters per candidate and 1,500 for the query (chosen on the dev split). LLMs decode greedily. The listwise prompt numbers the candidates and asks for a JSON array

of their numbers; candidates the model leaves out are appended in random order, and an unparsable answer falls back to a random order, never to the input order. The pointwise prompt asks for a 0–10 score, which we read as the expected value over the first generated token's probabilities.

4.2 Citation Labels

The queries are 174 papers from OpenAlex published in 2020 or later by seven institutions (MIT, Stanford, Harvard, UC Berkeley, the University of Michigan, Google and Google DeepMind), each citing 6–14 works; 168 have the required metadata, and a seeded split assigns 50 to dev and 118 to test. A query's positives are the references it cites that have a title and an abstract (3–14 per query). Its $2n$ negatives, for n positives, are drawn from the other queries' references published before the query and sharing no author with it, either uniformly (*random*) or among papers that share one of the query's top-5 OpenAlex concepts (*concept*). The concept condition is harder but matches on field rather than topic: 97.7% of its negatives share a concept with the query, more than the positives do (73.5%), but 68.7% share only a top-level field such as "Medicine". Candidates are shuffled per query and seed, and the order shown to each method is stored. Because MRR saturates near 1 for strong methods, R-precision is the primary metric; a random ranking scores 33.3%.

The full-corpus setting ranks each query against 11,967 works: every eligible positive plus every work cited by at least five earlier papers that an OpenAlex title search returned for any query. These citing papers also form the co-citation graph used in Section 8, built under five leakage rules, each covered by a unit test: (R1) a query's own reference list is never used; (R2) neither the query nor its duplicates act as citing papers; (R3) citing papers that duplicate any query, including earlier versions, are dropped; (R4) only citing papers dated before the query count, with year-only dates moved to the end of the year; and (R5) no query record acts as a citing paper.

4.3 Human Expertise Labels

In the gold-standard dataset of Stelmakh et al. [34], 58 researchers gave 477 ratings, on a 1–5 scale, of their expertise in reviewing 463 papers they had read, 3–10 papers each. Each researcher is represented by up to 20 of their own papers, in ten random profile samples. Lexical and dense methods score a rated paper by its maximum similarity to the profile papers, as our deployed system does; Qwen3-Reranker-4B uses each profile paper as a query and also takes the maximum; the LLMs see the profile titles and the candidate, with prompts that keep the citation-benchmark structure but ask about reviewing expertise. We report the authors' weighted Kendall-τ loss, for which lower is better and a constant ranking scores 0.5, averaged over profile samples; our evaluation reproduces their published baselines to two decimals. This setting ranks papers for one researcher, the reverse of our system's direction, and covers a subset of the methods: the three encoders other than EmbeddingGemma, both lexical methods, listwise and pointwise Llama, and Qwen3-Reranker-4B.

Table 2: R-precision (%) of listwise rerankers under three candidate orders, citation labels, test split, shuffle seed 0 (*random* / *concept* negatives). Qwen3-Embedding does not depend on order. A random ranking scores 33.3.

Model	Cited first	Shuffled	Cited last
Llama 3.2 3B	72.3 / 74.1	46.3 / 43.1	30.0 / 19.9
Llama 3.1 8B	73.9 / 69.0	67.1 / 60.6	89.4 / 85.9
Qwen3-Embedding-8B		87.0 / 80.1	

4.4 Statistics

We report 95% bootstrap confidence intervals over queries or researchers and compare methods with paired two-sided Wilcoxon signed-rank tests, Holm-corrected [11] within each pre-specified family, and report mean differences with every test [30]. On citation labels, order-sensitive LLM methods are averaged over five shuffle seeds. The bibliographic-coupling analysis on human labels was pre-registered, the LLM design on human labels was fixed in code before any LLM loss was computed, and the fusion analyses on human labels are post-hoc; we mark them as such.

5 RQ1: Candidate Order Changes the Answer

We show each listwise model the same candidates in three orders: cited papers first, the order of our first evaluation; shuffled; and the reverse of the first order, so that the cited papers come last. Prompt, text and decoding are those of Section 4.1; the comparison uses the 118 test queries of both benchmarks and shuffle seed 0. Table 2 gives the results.

The 3B model favours early candidates. Moving the cited papers from first to last costs it 42.3 R-precision points on *random* and 54.2 on *concept* negatives (paired Wilcoxon, $p \le 6 \cdot 10^{-12}$), and its MRR falls from 0.95 and 0.98 to 0.33 and 0.27, below the 0.56 of a random ranking on both benchmarks; R-precision falls to or below chance on both, significantly only on *concept*. With the cited papers first, its MRR exceeds the 0.84 that our first evaluation reported, so candidate order alone accounts for the LLM's score there; even in that order it stays below Qwen3-Embedding ($p \le 0.031$), and its apparent lead in the first evaluation also needed the indexing error F6 (Section 3).

The 8B model favours late candidates. Its R-precision rises by 15.5 and 16.9 points when the cited papers move to the end ($p < 10^{-4}$), and in that order it scores above Qwen3-Embedding on both benchmarks, significantly on *concept* ($p = 0.004$) but not on *random* ($p = 0.10$). The two models fail in different ways. In shuffled lists, 52.6% of the 3B model's rankings follow the order they were shown in (Spearman $\rho > 0.8$ between output and input position, over rankings with at least five candidates), against 0.9% of the 8B model's; the 8B model reverses the shown order more often (10.3% vs. 5.1%), and half of its rankings (51.5%) are close to order-independent ($|\rho| \le 0.3$). Order sensitivity therefore does not disappear with scale: a fixed order inflates the 3B model's score, and with the cited papers last it lifts the 8B model above dense retrieval. Figure 2 shows the same pattern without any labels: in shuffled lists the 3B model ranks the candidates shown in the first tenth of the list at a mean relative position of 0.24 and those in the last tenth at 0.73,

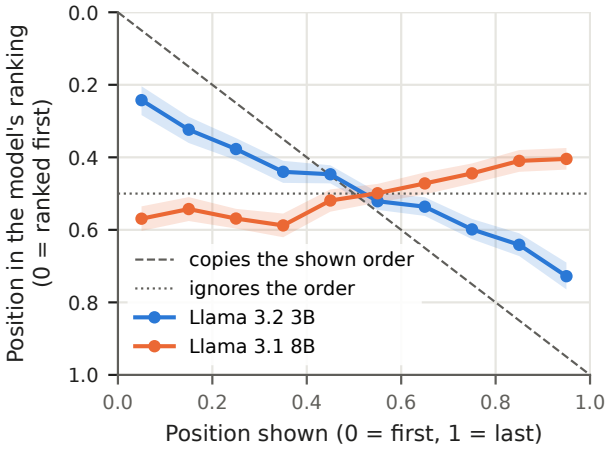


Figure 2: Where each model ranks a candidate, by the position it was shown at: mean relative position in the model's own output, for tenths of the shown list, with 95% bootstrap intervals over queries (shuffled lists, citation test split, both benchmarks, shuffle seed 0). No labels are used.

while the 8B model does the reverse, more weakly (0.57 and 0.40). Nor is the effect an artefact of incomplete outputs: on the queries where a model returns a complete ranking in all three orders, moving the cited papers from first to last changes R-precision by +32.9 and +41.9 points for the 3B model (36 and 26 queries) and by −12.0 and −12.1 for the 8B (79 and 85 queries; all $p \le 8.4 \cdot 10^{-3}$).

Both effects replicate on human labels (Figure 3, right). The 3B model's loss is 0.253 with the highest-rated papers first, 0.467 shuffled and 0.685 with them last (best- vs. worst-first, Holm $p = 1.4 \cdot 10^{-5}$). The 8B model's is 0.413, 0.388 and 0.253: it again does best when the best papers come last (worst- vs. best-first, 43 of 58 researchers, Holm $p = 7.7 \cdot 10^{-5}$), and its rankings correlate negatively with the shown order (mean Kendall $\tau = -0.20$ when shuffled). The direction of the bias is a property of the model, not of the label source.

6 RQ2: Which Method Wins Under Controlled Order?

Under shuffled order, Qwen3-Embedding-8B has the best point estimate of the methods we ran on both label sources, 38–43 R-precision points ahead of the 3B listwise reranker and 20.6 / 17.3 ahead of the 8B one (66.3 / 62.8 over five seeds, Holm $p \le 2 \cdot 10^{-14}$), while a 22M-parameter MiniLM stays within 4.3 points of it. On human labels the 8B model improves on the 3B (loss 0.388 vs. 0.467) but stays well behind Qwen3-Embedding (0.222; worse for 46 of 58 researchers, Holm $p = 4.6 \cdot 10^{-6}$). Figure 4 places the eight methods run on both label sources side by side. The two sources order them similarly (Kendall $\tau = 0.79$, 95% bootstrap interval [0.57, 0.86]; three of 28 pairs swap): the embeddings lead on both, and the listwise Llama models trail on both, the 3B one close to a constant ranking on human labels. The swaps involve the lexical methods and pointwise Llama, which is fourth on citation labels and sixth on human labels.

Table 3: Second stages and LLM scorers against Qwen3-Embedding-8B alone, as the change from the embedding. Pools (↑): R-precision (random / concept), reranking its top 10. Full corpus (↑): nDCG@10 (embedding 0.340), reranking its top 100. Human labels (↓): τ loss (embedding 0.222); here the LLM and the reranker score papers alone, and only the fusion uses the embedding. p: Holm-corrected Wilcoxon unless marked.

Setting	Second stage	Change
Pools	Llama 3B listwise, top 10	-13.0 / -10.4 ($p \le 3 \cdot 10^{-10}$)
Full corpus	Llama 3B pointwise	-0.164 ($p = 3.1 \cdot 10^{-11}$)
	fused ($\alpha = 0.1$)	-0.004 (n.s.)
	Qwen3-Reranker-4B	+0.002 (n.s.)
	fused ($\alpha = 0.8$)	+0.021 ($p = 0.22$)
Human labels	Llama 3B pointwise	+0.115 ($p = 3.9 \cdot 10^{-4}$)
	Qwen3-Reranker-4B	+0.046 ($p = 9.4 \cdot 10^{-3}$)
	fused (post-hoc)	+0.011 ($p = 2.0 \cdot 10^{-4}$)[a]

[a] A single uncorrected test. The weight is chosen leave-one-researcher-out on a flat optimum: in-sample, the best weight only ties the embedding (0.2223 both), so we read this as no gain rather than harm.

7 RQ3: Does a Second Stage Help?

Table 3 collects every second stage we tried on top of Qwen3-Embedding. In the cascade on the fixed pools, the embedding's top 10 already holds 93% and 89% of the cited papers, so the first stage leaves little to recover; listwise Llama reranking of those ten nonetheless lowers R-precision from 87.0 to 74.0 and from 80.1 to 69.7 ($p \le 3 \cdot 10^{-10}$). In the full corpus, where the embedding's top 100 is reranked, pointwise Llama alone halves nDCG@10, and fusing its score with the embedding's (weight chosen on dev) changes nothing. Qwen3-Reranker-4B, a model trained for reranking, ranks the top 100 better than the embedding on its own (within-list AUC 0.824 vs. 0.761) but improves nDCG@10 only when fused, by 0.021, which does not survive Holm correction; R-precision rises from 25.1 to 28.3 (Holm $p = 0.040$). On human labels the same reranker is worse than the embedding alone, and a post-hoc fusion with leave-one-researcher-out weights does not beat the embedding either. The only second-stage gain in this study therefore depends on the label source, and we do not claim it.

8 RQ4: Do Citation-Graph Signals Add Information?

Co-citation in the full corpus. Qwen3-Embedding finds 64.0% [58.7, 69.4] of the cited papers in its top 100, so 36% never reach a second stage. We replace the bottom 10% of that list with the papers most often co-cited with its top 5 [33], a seed-and-replace scheme in the style of corpus-graph expansion [14, 19], with seed and replacement sizes chosen on dev. Under the five leakage rules (Section 4.2) only 54.7% of test positives have any allowed citing paper; without the conservative date rule this would be 62.9%, and without the date rule at all 68.1%, so each rule removes real leaks,

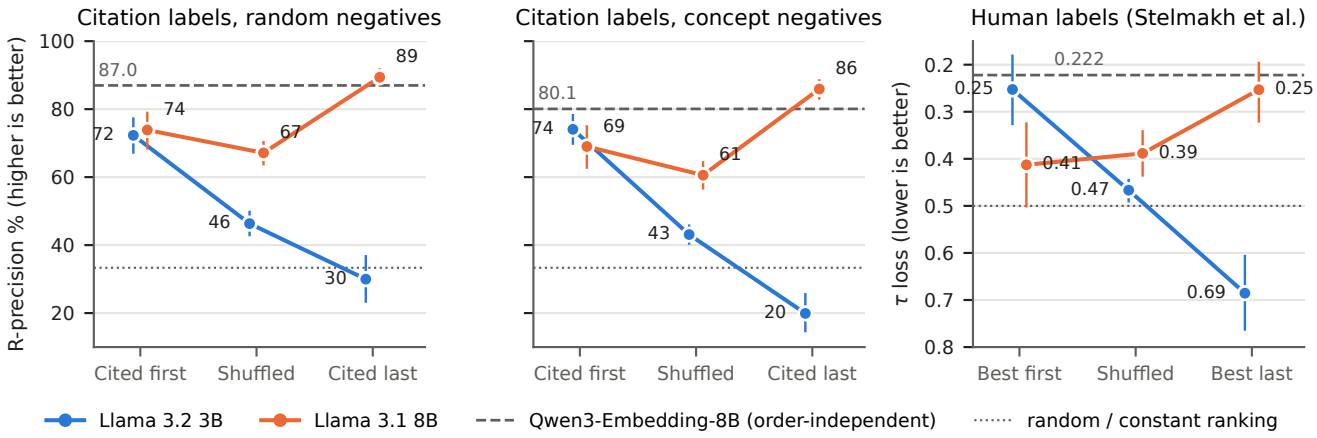


Figure 3: Listwise Llama 3.2 3B and 3.1 8B under three candidate orders, with 95% bootstrap intervals. Left and middle: R-precision on citation labels (test split, shuffle seed 0). Right: weighted Kendall-τ loss on human labels (five seeds), axis inverted so that up is better in every panel. The 3B model does best when the relevant papers come first, the 8B model when they come last.

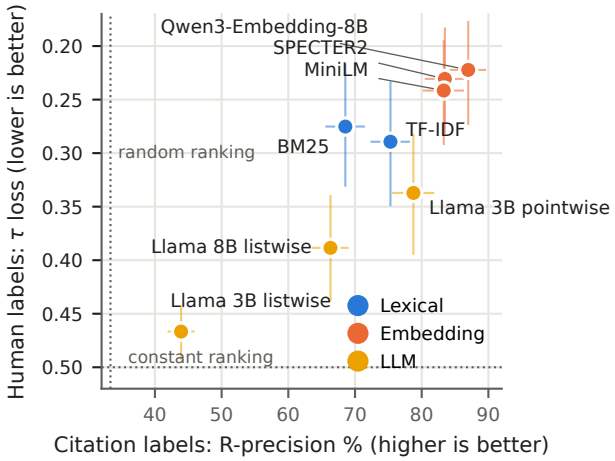


Figure 4: Citation labels (R-precision, *random* negatives, test split) against human labels (weighted Kendall-τ loss, lower is better), with 95% bootstrap intervals, for the eight methods run on both; LLM methods are averaged over five shuffle seeds. Up and to the right is better on both axes.

such as a 2012 record of a review that holds all ten references of a 2020 query. Table 4 shows the result. The expansion gains 0.8 points, not significantly, and little more than its controls: the same expansion over a degree-preserving random graph loses 0.5–0.9 points, and over a k-nearest-neighbour graph built from the embeddings themselves [15] it gains 0.3; only 20 of the 548 positives in the expanded lists (3.6%) came through the graph. A canary that adds each query's own references to the graph gains 5.8 points, so the method recovers papers when the signal is present; under leakage control, it is not.

Table 4: Citation-graph signals. Full corpus: change in recall@100 from Qwen3-Embedding alone (64.0), 95% paired bootstrap interval; co-citation is not significant (Holm $p = 1.0$), the canary is ($p = 1.6 \cdot 10^{-4}$). Human labels: τ loss (lower is better). p: Holm-corrected Wilcoxon against the dense model alone.

Setting	Signal	Result
Full corpus	Co-citation, all rules	+0.8 [−0.4, +2.1]
	Random graph (same degrees)	−0.5 to −0.9
	Embedding kNN graph	+0.3 [−0.6, +1.3]
	Canary (own references)	+5.8 [+3.6, +8.2]
Human	Coupling alone	0.299 [0.248, 0.348]
	Qwen3 / + coupling	0.222 / 0.212 ($p = 1.0$)
	SPECTER2 / + coupling	0.231 / 0.197 ($p = 8.3 \cdot 10^{-3}$)

Bibliographic coupling on human labels. Coupling scores a paper by the references it shares with a researcher's papers [13], a signal that exists as soon as a paper is published. In an analysis pre-registered before any score was computed, it carries real signal (loss 0.299, better than a constant ranking for 46 of 58 researchers, Holm $p = 4.2 \cdot 10^{-6}$) but trails every embedding, though not significantly. Fused with an embedding (weights chosen leave-one-researcher-out), it improves SPECTER2 significantly but not Qwen3-Embedding, the model the deployed system uses. On both label sources, the citation graph adds no significant gain to Qwen3-Embedding.

9 Discussion

Evaluate rerankers under several orders. A listwise reranker's score depends on the order it is shown, and the direction of that dependence differs between models. An evaluation that keeps one fixed order can therefore misstate a reranker's quality: cited papers

first inflated the 3B model in our first evaluation, and cited papers last would have put the 8B model above the embedding. Shuffling the candidates per query and seed, storing the order shown, and reporting the cited-first and cited-last conditions next to the shuffled one costs three runs and exposes the effect at once. Failure handling matters as well: a parse failure must fall back to a random order, never to the input order. Aggregating over permutations [36] reduces the bias, but multiplies the cost of a stage that, in our experiments, did not help.

Citation labels and human labels mostly agree. The two label sources order the eight methods they share similarly (Kendall τ = 0.79 [0.57, 0.86]), and every claim about position bias and about the embedding's lead holds on both. Two results do not transfer: pointwise Llama scores above the lexical baselines on citation labels (in point estimate) but below them on human labels, and fusing a trained reranker with the embedding helps on citation labels only. Citation-graph signals behave in between. When bibliographic coupling is strong, it is precise: the top decile of non-zero coupling scores has a mean rating of 4.50 and 88% of its papers are rated 4 or higher, and a researcher's top paper by coupling is rated as highly as the top paper by Qwen3-Embedding (4.39 vs. 4.37). But 30% of rated papers share no reference with the researcher's profile at all, so coupling is better used as a high-precision positive label than as a full ranking.

Implications for the deployed system. For our recommender these results favour a single embedding stage. We removed the LLM stage from the system's matching code; it cost about 0.35 s per pointwise call with the 3B model, and 7–9 s per listwise candidate list with the 3B model and about 18 s with the 8B, while an exact embedding search over the system's Boğaziçi works needs no model call beyond encoding the new paper. Because a 22M-parameter MiniLM stays within about four R-precision points of the 8B embedding model, a small encoder remains a defensible choice where the larger one cannot be hosted.

10 Limitations

Labels. Citations measure what a paper cites, not what a reader finds relevant; people also cite methods, tools and background. The human labels cover 58 researchers, almost all in computer science, and rank papers for one researcher, the reverse of our system's direction; on them, the LLMs see a researcher's profile titles while the encoders see titles and abstracts. Our *concept* negatives share a field with the query rather than its topic, so neither citation benchmark contains same-topic hard negatives. *Models.* We test two general Llama models (3.2 3B and 3.1 8B, both 4-bit), which differ in release as well as size, one trained reranker and one cross-encoder; larger and hosted LLMs, and rerankers fine-tuned for listwise ranking, may behave differently. On citation labels the order comparison uses one shuffle seed (the method comparison averages five), and the prompts on human labels were written for the task without tuning. The embedding models run without their recommended instruction prefixes. *Contamination.* Only one test query appeared after the Llama models' training cutoff, Qwen3-Reranker-4B was released after every query, and the human labels have been public since 2023, so we cannot rule out that some models have seen the papers or their labels [29].

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A Data Quality and Robustness

Abstracts. About 6% of OpenAlex "abstracts" are not abstracts (author lists, citation strings or another paper's title); dropping the affected test queries moves no method by more than 1.1 points. *Seeds.* Over five shuffle seeds, the standard deviation of each LLM method's R-precision is 0.8–2.0 points (listwise 8B: 0.9 / 2.0). *Output failures.* Listwise 3B returns a partial ranking for 38% of full-list calls (9–42 candidates); its sliding-window variant, at most 10 candidates per call, fails on about 2% of queries and still scores 55.6 / 52.2, far below the embeddings. The 8B model returns partial rankings for 7–19% of queries per condition, mostly as complete answers that leave out some candidates rather than as truncated ones. On human labels, lists are shorter and almost every call succeeds. *Coverage.* Adding references found by title search raises profile coverage from 80.3% to 85.2% and changes no coupling conclusion.